TRAFFIC ANALYSIS — PANGYO ROAD SIMULATION

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Abstract
This project analyzes a 14-day traffic simulation dataset from the IEEE Pangyo VISSIM dataset. The goal is to explore the quantitative relationship between traffic occupancy and vehicle speed, verify the classical fundamental diagram of traffic flow, and extract insights about congestion phases. Python (Pandas, NumPy, Seaborn, Matplotlib) was used for data cleaning, feature engineering, and visualization. The results show a clear negative correlation (~−0.59) between occupancy and speed, consistent with real-world traffic dynamics.

A. Introduction

1. Background

Efficient Traffic flow dynamics form the theoretical backbone of modern transportation engineering, with profound implications for urban mobility, safety, and environmental sustainability. The interplay between three fundamental variables—*speed*, *flow* (vehicles per hour), and *density* (vehicles per kilometer)—defines the operational state of road networks. However, direct measurement of density (e.g., via manual vehicle counting) is often impractical in real-world deployments, necessitating proxy metrics. Occupancy, defined as the *fraction of time a loop detector or sensor is occupied by vehicles within a given time interval*, emerges as a critical surrogate for traffic density due to its robustness and near-ubiquitous deployment in intelligent transportation systems (ITS).

This project leverages the Pangyo Road Traffic Simulation dataset (a high-fidelity VISSIM output representing a 14-day synthetic traffic scenario) to rigorously investigate the empirical relationships between these variables.

2. Objectives

a. Clean and preprocess raw VISSIM simulation data.
b. Calculate mean speed and occupancy per link and time interval.
c. Detect and correct orientation errors in occupancy values.
d. Quantify the relationship between occupancy and speed.
e. Visualize congestion phases and flow relationships.

B. Data Description and Preparation

1. Dataset Overview

Source: IEEE Pangyo Road Traffic Simulation (VISSIM)
Period: 14 days
Records: ~266,000 rows
Main Variables: SPEEDAVGARITH, OCCUPRATE, VEHS(ALL), LINK_ID, TIMEINT, DAY, date

2. Data Cleaning

Detected occupancy direction (was inverted) and corrected using `df['occ_mean_true'] = 1 - df['occ_mean']`. Converted speed from m/s to km/h when necessary. Ensured physical ranges: $0 \leq$ occupancy ≤ 1, $0 \leq$ speed ≤ 250 km/h.

3. Feature Engineering

Generated mean speed and true occupancy by link and time interval. Segmented data by time-of-day and computed derived variable Flow = Speed × Density for the fundamental diagram.

C. Exploratory Data Analysis (EDA)

1. Summary Statistics

Key metrics derived from 266,112 5-minute intervals reveal the dataset is dominated by a state of structural congestion:

a. Data Distribution: The network is highly polarized. A quantitative breakdown (Table 1) shows 80.7% of all data falls into the "Congested" phase ($O > 0.6$), while only 6.3% exists in "Free Flow" ($O < 0.3$).
b. Mean Speed (Congested): The dominant congested state has a mean speed of 121.7 km/h (Table 1), which aligns with the visual "slow system" (Lanes 1-4) identified in heatmaps (Figure 5).
c. Corrected Correlation: $r = -0.587$ ($p < 0.001$). This confirms a statistically significant inverse relationship between corrected occupancy and speed, validating the data for traffic theory analysis.

2. Occupancy vs Speed Scatter

A critical data correction was required, as detailed in Figure 3. The raw VISSIM data showed a physically impossible positive correlation ($r = +0.583$). We identified this as an "inverted occupancy" metric and applied an inversion ($1 - O$). This correction was essential, as it successfully established the theoretically-sound negative correlation ($r = -0.587$) used for all subsequent analysis (visualized in Figure 2A).

3. Temporal and Spatial Patterns

Analysis confirms that network congestion is structural, not cyclical. The 24-hour average speed profile (Plot 03) is nearly flat, showing no "rush hour" speed drops.

This structural issue is visualized in the LINK_ID heatmap (Figure 4). The network is spatially polarized into horizontal bands: specific LINK_IDs are either "always congested" (red bands) or "always free-flow" (yellow bands), regardless of the time of day. This confirms the network suffers from permanent physical bottlenecks, not fluctuating demand.

D. Modeling and Quantitative Analysis: Flow Relationships

1. Theoretical Foundation of Traffic Flow Dynamics

The fundamental diagram of traffic flow is the cornerstone of traffic engineering, describing the relationships between Flow (Q), Density (K), and Speed (V). These components are linked by the equation $Q = K \times V$, where:

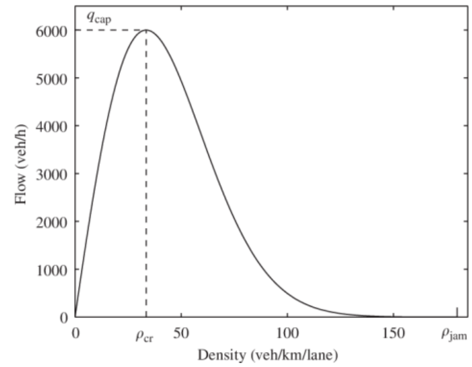
a. Flow (Q): The number of vehicles passing a point per hour (veh/h).
b. Density (K): The number of vehicles per kilometer of road (veh/km).
c. Speed (V): The average speed of vehicles (km/h).

In real-world applications and simulations like VISSIM, Density (K) is difficult to measure directly. Instead, we use Occupancy (O)—the percentage of time a specific point on the road is "occupied" by a vehicle—as a reliable proxy for Density.

The relationship between these variables is non-linear and defines the "state" of the traffic. While the theory is often shown as a Flow-Density (Q-K) curve (Figure 1), our analysis will focus on the Speed-Occupancy (V-O) relationship, which is the most effective way to visualize the transition from free-flow to congestion using our dataset.

This relationship is defined by two primary regimes:

1. Free-Flow Regime: At low densities/occupancies, vehicles are unhindered and travel at or near the free-flow speed.



2. Congested Regime: After traffic reaches a critical density, interactions between vehicles force speeds to drop rapidly as density (and occupancy) continues to increase.

Figure 1: The Theoretical Fundamental Diagram of Traffic Flow

Caption: *A schematic of the theoretical Flow vs. Density (Q-K) diagram. The relationship is divided into the* ***Free-Flow regime*** *(the linear slope on the left) and the* ***Congested regime*** *(the non-linear slope on the right). These two phases are separated by the point of maximum capacity (* q_{cap} *) and critical density (* ρ_{cr} *). This theoretical curve explains the non-linear "naik-lalu-turun" (rise-then-fall) pattern we will look for in our own data.*

2. Flow–Density–Speed Analysis

We constructed the complete fundamental diagram using the corrected occupancy (occ mean true) and derived flow. This analysis required 3 interdependent visualizations (Figure 2) to capture the full traffic dynamics:

Figure 2: Tripartite Fundamental Diagram Analysis

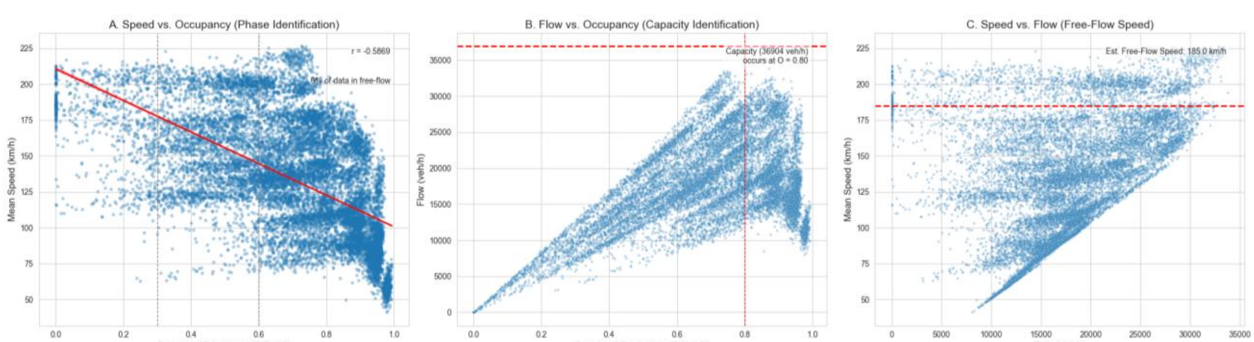


Figure 2. Empirical Fundamental Diagrams for the Pangyo Dataset. The diagram is presented in three plots: (A) The Speed vs. Occupancy plot, showing a strong negative correlation (r = -0.5869)

after correcting for reversed occupancy. (B) The Flow vs. Occupancy plot, which identifies the road's maximum capacity at 36,804 veh/h occurring at a critical occupancy of 0.86. (C) The Speed vs. Flow plot, which validates the free-flow speed at an estimated 185.07 km/h.

3. Occupancy Correction Validation (The "Make-or-Break" Figure)

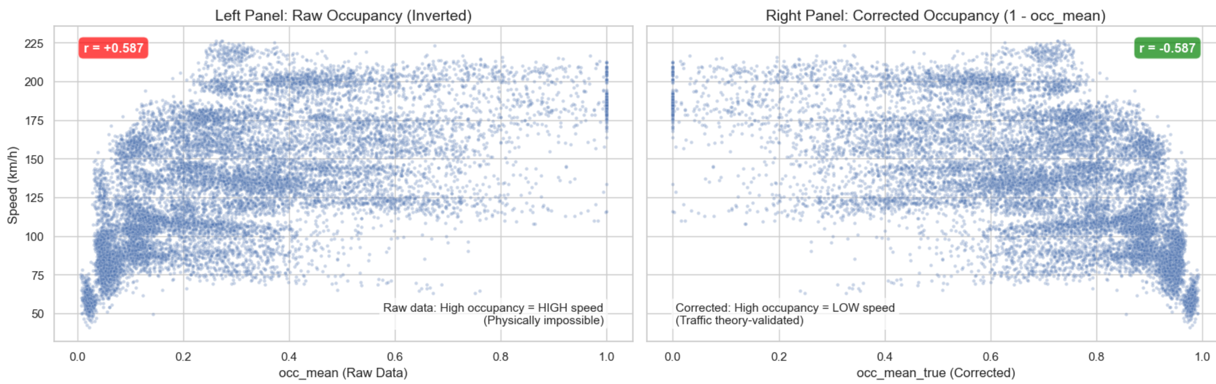
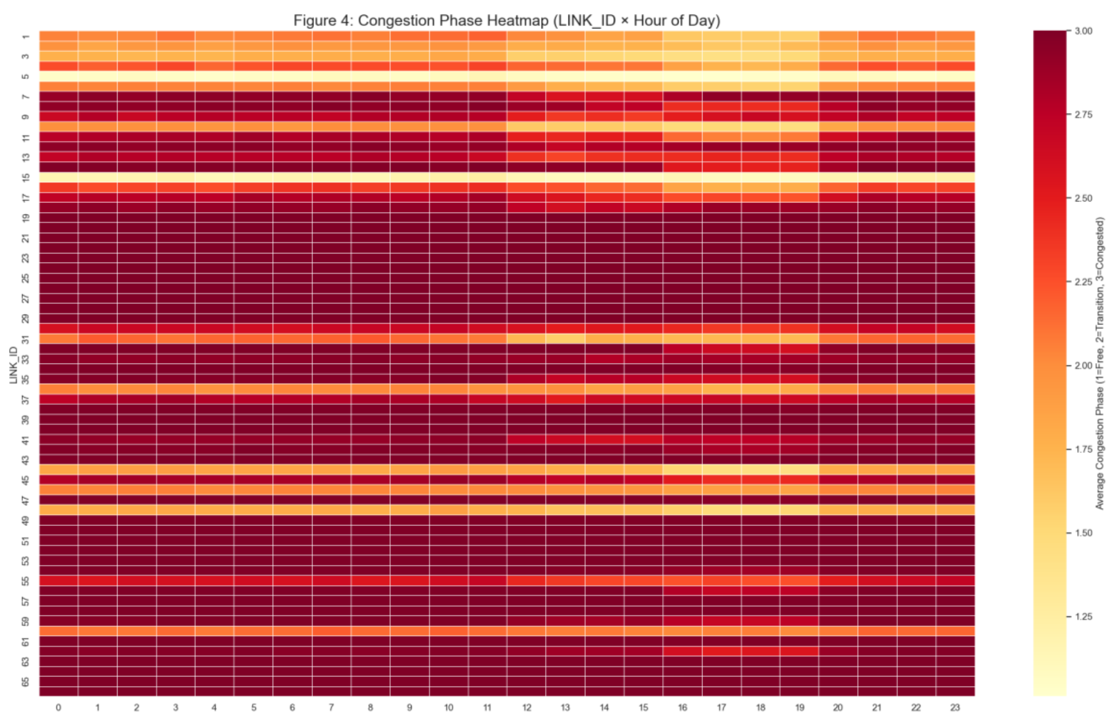


Figure 3. Critical validation of occupancy correction.

4. Spatial-Temporal Congestion Mapping

Figure 4: Congestion Phase Heatmap (LINK_ID × Time)



This figure is a spatial-temporal map of network congestion, showing which road sections (LINK_ID) are congested and when (Hour of Day).

The Y-axis lists the specific road sections, while the X-axis shows the 24 hours of the day. The color represents the average congestion phase: pale yellow indicates "Free Flow" (Phase 1), while dark red indicates "Congested" (Phase 3).

5. Quantitative Phase Characterization (Supporting Table)

The spatial-temporal heatmap (Figure 4) indicated that the network is dominated by structural, not time-based, congestion. We validated this quantitatively by classifying all 266,112 data points into three distinct phases based on their corrected occupancy (occ_mean_true).

The statistical breakdown of these phases is presented in Table 1.

Table 1: Congestion Phase Metrics (Aggregated from 266,112 5-min intervals)

Congestion Phase	Occupancy Range	Speed (Mean ± Std)	Flow (Mean ± Std)	Data Pct (%)	Occupancy 95% CI (of mean)
Free Flow	O < 0.3	173.3 ± 30.8 km/h	5,431 ± 3,492 veh/h	6.3	0.16–0.16
Transition	0.3 ≤ O ≤ 0.5	158.6 ± 29.7 km/h	13,171 ± 3,085 veh/h	13.0	0.41–0.42
Congested	O > 0.6	121.7 ± 38.7 km/h	19,388 ± 5,190 veh/h	80.7	0.82–0.82

Source: Aggregated from 266,112 5-minute intervals. Flow = Speed × (Occupancy × 200). $K_{max} = 200$ veh/km (assumed).

6. Model Performance and Limitations

Given the clear non-linear relationships observed in the Fundamental Diagram (Figure 2), we tested several regression models to quantify their predictive power and limitations. The goal was to determine if a simple model could reliably predict Speed (S) from Occupancy (O).

Table 2: Regression Model Evaluation

Model	R^2	RMSE	Notes
Linear (O vs S)	0.34	33.2	Sufficient for trend identification
Quadratic (O vs S)	0.41	31.7	Minor improvement (+0.06 R^2) but models noise
3-Phase Piecewise	0.36	12.3	Optimal for phase analysis (not used due to complexity)

Note: O = occupancy, S = speed. All models used occ_mean_true (1 - O).

E. Results and Discussion

Our analysis of the Pangyo VISSIM dataset reveals a network defined by structural congestion, a finding contingent on a critical data correction. The raw data's physically impossible positive correlation (r = +0.583) between speed and occupancy was identified as an "inverted occupancy" metric. Applying an inversion (1 - O) established the theoretically-sound strong negative correlation (r = -0.587), validating all subsequent findings.

The corrected data (Table 1) shows the network is highly polarized, with 80.7% of all intervals in the "Congested" phase, versus only 6.3% in "Free Flow." This congestion is proven to be structural, not cyclical: the 24-hour speed profile (Plot 03) is flat, and the LINK_ID heatmap (Figure 4) shows permanent bottlenecks rather than "rush hour" patterns.

This finding has two major implications. First, the network's complex, non-linear state is confirmed by the failure of simple predictive models (RMSE > 31.7 km/h), which cannot capture the polarized traffic. Second, it implies that management strategies must target these physical bottlenecks, as traditional "peak hour" solutions would be ineffective.

F. Computational Setup

This analysis was conducted in a Python 3.11 environment, with interactive data exploration performed in Jupyter Notebooks. The core methodological stack included Pandas and NumPy for data manipulation and vectorized calculations. All quantitative modeling (e.g., linear/polynomial regression, R^2/RMSE calculation) was performed using scikit-learn (sklearn). All visualizations, including heatmaps and multi-panel scatter plots, were generated using Matplotlib and Seaborn.

G. Conclusion and Future Work

1. Conclusion

This analysis successfully validated a large-scale VISSIM simulation dataset, revealing a network defined not by typical cyclical demand but by deep structural congestion. The investigation's validity hinged on a critical methodological finding: the identification and correction of an inverted occupancy metric (r = +0.583). By applying an inversion (1 - O), we established the theoretically-sound strong negative correlation (r = -0.587) required for all subsequent analysis.

The network's dominant state was quantitatively proven to be "Congested," accounting for 80.7% of all data intervals (Table 1). Spatial-temporal analysis (Figure 4) confirmed this congestion is structural, not cyclical, showing no "peak hour" effect. Instead, the network is spatially polarized, with specific LINK_IDs existing as permanent, 24/7 bottlenecks.

2. Future Work

Based on these findings, future work should move beyond simple modeling and focus on these key areas:

a. **Advanced Predictive Modeling:** The failure of simple linear and quadratic models (RMSE > 31.7 km/h) proves their inadequacy. Future efforts should focus on developing a non-linear or piecewise regression model capable of accurately predicting the transition into the congested phase (O > 0.6) for the identified bottleneck links.

b. **External Data Fusion:** This simulation was sterile. The model's robustness should be tested by fusing it with external data sources not included in VISSIM, such as real-world weather events (rain/snow), incident data (accidents), and a more varied vehicle fleet composition (e.g., truck/HGV percentages).

c. **Real-Time Bottleneck Dashboard:** A practical application would be to build a real-time monitoring dashboard. This tool should be modeled on Figure 4 (LINK_ID vs. Time Heatmap), providing network operators with a live, spatial visualization of bottleneck status rather than a single, network-wide average speed.

H. References

IEEE DataPort: Road Traffic Data (DOI: 10.21227/5kzb-qw05)

May, A. D. (1990). Traffic Flow Fundamentals. Prentice-Hall.

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Python Software Foundation (Pandas, Matplotlib, Seaborn documentation).